Universidade de Brasília
Statistics Department

Multivariate analysis of factor determining the performance of players in the Brasileirão Série A - 2022

Francisco Iago dos Reis Ferreira

Project presented to the Department of Statistics at the University of Brasilia as part of the requirements for obtaining a Bachelor's degree in Statistics.

Brasília
2023

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Thanks to

It's been a long journey, and I would like to thank all the people who have contributed to this work.

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Resumo

O Campeonato Brasileiro, também conhecido como Brasileirão, é a principal competição de futebol profissional no Brasil. sendo disputado todos os anos por clubes que representam diversos estados de todo o país. Adotado em 2003, o formato de pontos corridos é disputado por 20 clubes que se enfrentam em jogos de ida e volta durante 38 rodadas, a equipe que fizer a maior pontuação se sagra como a campeã daquela edição. Esse trabalho teve como objetivo empregar técnicas de Análise Multivariada com o intuito de identificar os principais fatores que influenciam o desempenho dos clubes participantes do Campeonato a partir de dados relativos aos jogadores que integram os times na competição disputada em 2022.

Palavras-chaves: Brasileirão, Desempenho, Análise Fatorial, Análise de Componentes Principais.

Abstract

The Brazilian Championship, also known as the Brasileirão, is the main professional soccer competition in Brazil and is contested every year by clubs representing various states across the country. Adopted in 2003, the straight points format is played by 20 clubs who face each other in back-to-back matches over 38 rounds, with the team that scores the highest being crowned champion of that edition. The aim of this study was to use Multivariate Analysis techniques to identify the main factors influencing the performance of the clubs taking part in the Championship, using data on the players who make up the teams in the competition in 2022.

Keywords: Brasileirão, Performance, Factor Analysis, Principal Component Analysis.

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1 Chapter 1 - Introduction

In 2002, Billy Beane, general manager of the *Oakland Athletics* baseball team, challenged everything that had been believed about competitive sport until then by adopting advanced statistical techniques to evaluate player performance. In contrast to the criteria used by scouts at the time, such as the ability to run, throw, receive the ball, hit and hit with power, Billy and Paul dePodesta (by then a young club executive) were able to conclude that not all the important characteristics in a baseball player are of equal importance. Running speed, defensive ability and brute strength tended to be overrated characteristics.

They were able to show that characteristics such as batting percentage, on-base percentage, base control ability and other statistical metrics were more reliable indicators of a player's performance than traditional evaluations. This approach revealed valuable information about a player's effectiveness, allowing talents often undervalued by other teams to be identified.

With a limited budget, being the team with the third-lowest payroll in the league at the time, but with an authentic statistical approach, the Oakland Athletics broke the record for consecutive wins in a season (LEWIS, 2004). The impact of this approach paved the way for a new era in sport, going beyond baseball and inspiring many different sports. The use of statistical techniques led to a change in the way players' performance was evaluated and understood.

In the context of soccer, it is imperative to consider the advances in the information contained in the data. According to Anderson and Sally (2013), numbers have the power to challenge pre-established concepts and dismantle long-held beliefs in sport. They offer an understanding of the game that has never been seen before, providing a counterpoint to traditional assumptions.

An example of how data can influence decision-making is the contract renewal made by Manchester City player Kevin de Bruyne[1]. At the time, the player managed to get a salary extension of around 80 million euros, using only data analysis instead of an agent[2].

In Brazil, the Brasileirão Serie A is a competition that moves millions of reais, instigating fans, investors and bettors every year. With the growing advance of techno-

[1] ⟨https://www.transfermarkt.com.br/kevin-de-bruyne/profil/spieler/88755⟩, accessed on 15/04/2023
[2] ⟨https://www.uol.com.br/esporte/futebol/ultimas-noticias/2021/04/08/ big-data-e-sem-agente-como-de-bruyne-renovou-com-o-city.htm⟩, accessed on 15/04/2023

logy, the use of statistical techniques has been of immense importance in assessing and understanding the performance of players during the competition. On the national stage, for example, teams such as Flamengo[3] and Palmeiras[4] use performance analysis to make administrative and sporting decisions.

This study aims to analyze, using Multivariate Analysis techniques, the performance of players in the first division of the 2022 Campeonato Brasileiro Série A, as it allows for a more accurate and complete evaluation, as well as identifying patterns and trends that would not be found through simpler analyses. Through this analysis, we hope to identify the characteristics that most contributed to the players' performance during the championship.

This approach applied to the context of the Brazilian championship can be useful for coaches, bettors, performance analysts, investors and sponsors, as well as soccer enthusiasts, who will be able to make better decisions in the sporting sphere using information equipped with statistical analysis.

[3] ⟨https://encurtador.com.br/rALOV⟩, accessed on 10/04/2023
[4] ⟨https://encurtador.com.br/doyMZ⟩, accessed on 10/04/2023

1.1 General Objective

The aim of this study was to use Multivariate Analysis techniques to identify the main factors influencing the performance of the clubs participating in the Brazilian Serie A Championship in 2022. Initially, the analysis was conducted for all 20 clubs, followed by a comparison between the performance of the clubs that reached the Libertadores group stage and those that were relegated to the second division. The line positions of the players in these teams were taken into account.

1.1.1 Specific Objectives

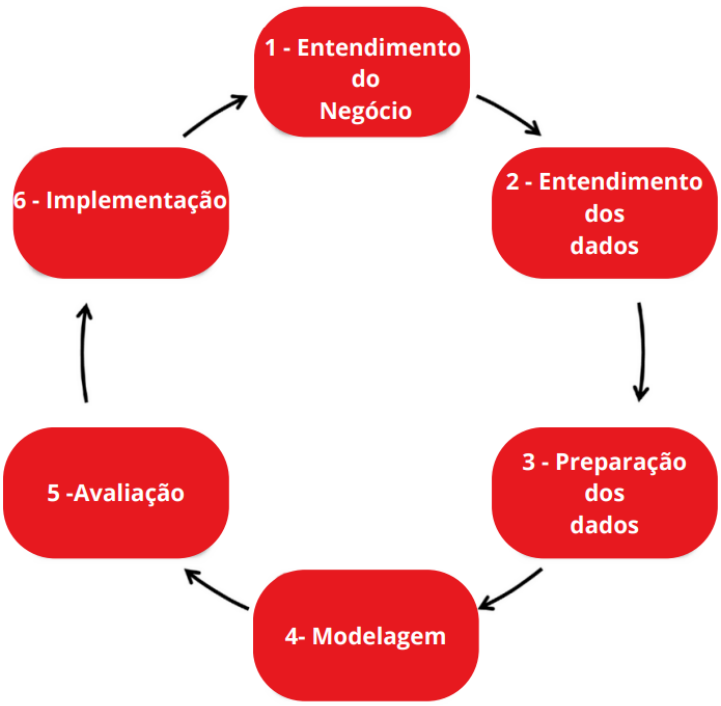
To achieve the general objective, the following steps must be taken:

- Collect and prepare the performance data of the players who took part in the championship;
- Perform an exploratory analysis of the data to identify possible missing values and inconsistencies;
- Apply dimensionality reduction techniques to select the most relevant variables;
- Validate the results obtained;
- Interpreting the results obtained.

1.2 Methodology

This work was based on the CRISP-DM methodology, with the aim of organizing the work in an efficient and standardized manner[5].

Figura 1: CRISP DM Methodology



Source: Adapted by the author (*Smart Vision*)

From Figure 1, we can organize the work as follows:

- **Phase 1 - Understanding the problem:** In the initial stage, data will be collected and organized.
- **Phase 2 - Understanding the data:** In this stage, missing and inconsistent data will be properly identified and processed.
- **Phase 3 - Data preparation:** After the previous stage, the data will be prepared for analysis, which includes standardizing the data and selecting the most relevant variables.
- **Phase 4 - Modeling:** Multivariate analysis methods will be used. We will also carry out tests to validate the results obtained.

[5] ⟨https://www.sv-europe.com/crisp-dm-methodology/⟩, accessed on 10/09/2023

- **Phase 5 - Evaluation:** The results will be evaluated in relation to the objectives of this work. If necessary, additional analyses will be carried out to obtain more accurate results.
- **Phase 6 - Conclusion:** The results obtained will be discussed and validated.

2 Chapter 2 - Statistical Method

This chapter describes the statistical analysis methods used in the research. These methods were selected due to their relevance within the context of the data collected and will contribute to achieving the objectives described in the previous section.

2.1 Multivariate Analysis

Multivariate analysis is a set of statistical methods used when multiple variables are measured simultaneously in each sample element. Generally, these are variables that are correlated with each other, and the greater the number, the more complex the analysis becomes. The aim is to simplify the interpretation of the phenomenon being studied. It can be divided into two main categories: exploratory techniques which seek to simplify the variability structure of the data by synthesizing the variables, and inference techniques. One of the common methods for simplifying the structure of variability is dimensionality reduction, which aims to reduce the number of original variables while retaining the essential information for analysis.

2.2 Principal Component Analysis

According to Jolliffe (2002), the essence of the Principal Component Analysis technique is to reduce the complexity of a data set containing numerous variables (P). This reduction seeks to preserve the variability of the original data as much as possible. This type of simplification can be achieved by transforming the data into a new set of variables called principal components (k). These components are not correlated with each other and are organized so that the first components can retain most of the variation present in the original set of data.

The mathematical development expressed in section 2.2 was taken from Johnson's work(2002).

The main objective of this method is to explain the variance/covariance structure of the data through linear combinations of the variables, where the P variables are retained in k components. We can write these components as linear combinations of the P variables $X_1, X_2, ..., X_p$. Geometrically, these combinations represent the selection of a new coordinate system obtained by rotating the original system with $X_1, X_2, ..., X_p$ as the coordinate axis; these axes represent the directions with maximum variability and

provide us with a more parsimonious description of the covariance structure.

Let the random vector **X'** $= [X_1, X_2, ..., X_p]$ with covariance matrix Σ have eigenvectors $\lambda_1 \geq \lambda_2 \geq ... \geq \lambda_p = 0$

Consider the linear combinations:

$$Y_1 = a_1'X = a_{11}X_1 + a_{12}X_2 + ... + a_{1p}X_p \tag{2.2.1}$$
$$Y_2 = a_2'X = a_{21}X_1 + a_{22}X_2 + ... + a_{2p}X_p \tag{2.2.2}$$
$$\vdots \tag{2.2.3}$$
$$Y_p = a_p'X = a_{p1}X_1 + a_{p2}X_2 + ... + a_{pp}X_p \tag{2.2.4}$$

which can be written as

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_p \end{bmatrix}_{p\times 1} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1p} \\ a_{21} & a_{22} & \cdots & a_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ a_{p1} & a_{p2} & \cdots & a_{pp} \end{bmatrix}_{p\times p} \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{bmatrix}_{p\times 1}$$

where

$$E(Y_i) = E(a_i'\mathbf{X}) \tag{2.2.5}$$
$$E(Y_i) = a_i'E(\mathbf{X}) \tag{2.2.6}$$
$$E(Y_i) = a_i'\mu \tag{2.2.7}$$

$$Var(Y_i) = V(a_i'\mathbf{X}) \tag{2.2.8}$$
$$Var(Y_i) = a_i'V(\mathbf{X})a_i \tag{2.2.9}$$
$$Var(Y_i) = a_i'\Sigma a_i, \quad i = 1, 2, ..., p \tag{2.2.10}$$

and

$$Cov(Y_i, Y_k) = V(a_i'\mathbf{X}, a_k'\mathbf{X}) \tag{2.2.11}$$
$$Cov(Y_i, Y_k) = a_i'\Sigma a_k, \quad i, k = 1, 2, ..., p \tag{2.2.12}$$

These are uncorrelated linear combinations of $Y_1, Y_2, ..., Y_p$ in which the variance is maximum. Where, we can define from this:

- The 1st principal component will be the linear combination of $a_1'X$ that maximizes $V(a_1'X)$ subject to the constraint $a_1'a_1 = 1$
- The 2nd principal component will be the linear combination of $a_2'X$ that maximizes $V(a_2'X)$ subject to the constraint $a_2'a_2 = 1$ and $Cov(a_1'X, a_2'X) = 0$

$$\vdots$$

- The i-th principal component will be the linear combination of $a_i'X$ that maximizes $V(a_i'X)$ subject to the constraint $a_i'a_k = 1$ and $Cov(a_i'X, a_k'X) = 0$ for $k < i$

Let $\boldsymbol{\Sigma}$ be the covariance matrix associated with the random vector $\boldsymbol{X' = [X_1, X_2, ..., X_p]}$, with eigenvalue-autovector pairs $(\lambda_1, e_1), (\lambda_2, e_2), ..., (\lambda_p, e_p)$, where $\lambda_1 \geq \lambda_2 \geq ... \geq \lambda_p \geq 0$, then the ith principal component is given by

$$Y_i \approx e_i'X = e_i1X + e_i2X + ... + e_ipX \quad i = 1, 2, ..., p \tag{2.2.13}$$

with

$$Var(Y_i) = e_i'\Sigma e_i = \lambda_i \quad i = 1, 2, ..., p \tag{2.2.14}$$

$$Cov(Y_i, Y_k) = e_i'\Sigma e_i = 0, \quad i \neq k \tag{2.2.15}$$

That is, $\boldsymbol{X' = [X_1, X_2, ..., X_p]}$ with covariance matrix Σ, and with eigenvalue-autovector pairs $(\lambda_1, e_1), (\lambda_2, e_2), ..., (\lambda_p, e_p)$, where $\lambda_1 \geq \lambda_2 \geq ... \geq \lambda_p \geq 0$, take $Y_i \approx e_i'X = e_i1X + e_i2X + ... + e_ipX \quad i = 1, 2, ..., p$, as the principal components. Then

$$\sigma_{11} + \sigma_{22} + ... + \sigma_{pp} = \sum_{i=1}^{p} Var(X_i) = \lambda_1 + \lambda_2 + ... + \lambda_p = \sum_{i=1}^{p} Var(Y_i) \tag{2.2.16}$$

If $Y_1 = e_1'X = Y_2 = e_2'X, ..., Y_p = e_p'X$ are the principal components obtained from the Σ covariance matrix, then

$$\rho_{Y_i X_k} = \frac{e_{ik}\sqrt{\lambda_i}}{\sqrt{\sigma_{kk}}} \quad i, k = 1, 2, ..., p \tag{2.2.17}$$

are the correlation coefficients between the components (Y_i) and the variables (X_k). And $(\lambda_1, e_1), (\lambda_2, e_2), ..., (\lambda_p, e_p)$ are the eigenvalue-autovector pairs for the Σ matrix. Alternatively, it is also possible to obtain the total proportion of the k-th principal component using the relation:

$$\frac{\lambda_k}{\lambda_1 + \lambda_2 + ... + \lambda_p} \quad i, k = 1, 2, ..., p \tag{2.2.18}$$

2.3 Related work

Billy Beane's work was a milestone when it came to statistical analysis in sport. The insertion of statistics into sport not only revolutionized the way baseball teams approached their strategies, but also served as an inspiration for various sports. In this sense, some of the works I'm going to mention aim to explore this relationship.

In Guimarães, Marques and Neto (2017), a study was carried out with the aim of evaluating the performance of soccer players from the German, English, Spanish, Italian and Brazilian leagues. They considered 91 attributes measured per player over the last three years in each of the leagues. The focus was on players who played as left-sided attacking midfielders. The insertion of statistics into sport not only revolutionized the way baseball teams approached their strategies, but also served as an inspiration for various sports. In this sense, some of the works I'm going to mention aim to explore this relationship.

Still in the area of soccer performance, Abrahão et al. (2020), verified the applicability of Factor Analysis by crossing variables from tests in a protocol for detecting athletic talent in soccer, which included anthropometric measurements, flexibility and strength. The sample involved 38 players from the under-19 category of Cuiabá Esporte Clube. The tests carried out included the iSports protocol for detecting sporting talent, anthropometric assessments (body mass, height, Body Mass Index - BMI, Waist and Hip Ratio - WHR), a flexibility test (Sit and Reach), a manual strength test (*Hand Grip*) and a vertical impulse test.

But there are also studies that have used simulators as an object of investigation, and Lima (2022) does this very well in this study. This analysis explores the growing role of computational techniques and resources in data analysis in the context of soccer, highlighting the usefulness of *Football Manager*[6] and specialized websites such as *Transfermarkt*. The aim was to identify relationships between footballers' indicators that suggest the development of their potential abilities. With a sample of 1309 players aged between 15 and 20, collected in the 2014 FM Scout, the study uses data mining techniques with the software Weka[7]. The results point to the relevance of investing in players under the age of 18, indicating that the use of *Football Manager* can be a valuable complementary tool to assist managers in decisions related to identifying and signing young talent.

In basketball, we can also see how statistics are crucial to understanding a team's

[6] ⟨https://www.footballmanager.com/⟩, accessed on 07/10/2023
[7] ⟨https://www.cs.waikato.ac.nz/~ml/weka/⟩, accessed on 07/10/2023

performance. Meneses, Luiz and Almeida 2016, examined the evolution of Brazilian basketball based on the performance indices of the teams over three seasons of Novo Basquete Brasil (from 2009 to 2012) and determined the reference values of the main DIs. The data was collected from the National Basketball League website. A total of 726 games were analyzed using one-way *ANOVA*. The 2011-2012 season stood out for having fewer three-point shots and more two-point shots, an improvement in free throws and a reduction in offensive rebounds. The percentiles indicate a high level of demand on the teams' performance for the IDs analyzed, and a trend of change in the teams' style of play has been observed over the last three seasons.

3 Chapter 3 - Results

3.1 Brazilian Football Championship 2022 - Serie A

Popularly known as the Brasileirão, this year saw the 67th edition of Brazil's top soccer division. It was played between April 9 and November 13, following the straight points format established in 2003, with 20 clubs competing in the championship since 2006. This year's champions were Palmeiras (SP), who won their 10th title after Internacional (RS) lost 1-0 to América Mineiro (MG).

At the bottom of the table, Juventude(RS) were the first club relegated to the 2023 Série B, after being beaten 1-0 by Atlético Mineiro(MG) on matchday 34. Avaí (SC), Ceará (CE) and Atlético Goianiense (GO) were the other clubs to be relegated soon after.

Palmeiras was the team with the best attacking performance, scoring 66 goals, and also had the most solid defense, conceding only 27 goals. The worst attack and worst defense belonged to relegated Juventude, with 29 and 69 goals respectively. Germán Cana, a player from Fluminense (RJ), won the title of top scorer in the league by scoring 26 goals, while Gustavo Scarpa, from Palmeiras, was the leader in assists, totaling 13.

Figure 2 shows the final league standings. The clubs ranked up to 14th took part in South America's international competitions. From Palmeiras to Athlético Paranaense (PR), they secured their places in the Libertadores group stage, while Atlético Mineiro and Fortaleza (CE) qualified for the Libertadores pre-qualifying stage. Between 9th and 14th place, the clubs secured their places in the Copa Sudamericana. Notably, Coritiba (PR) and Cuiabá (MT) were the only clubs that didn't have the opportunity to compete in any international competition the following year.

Figura 2: Table after Round 38

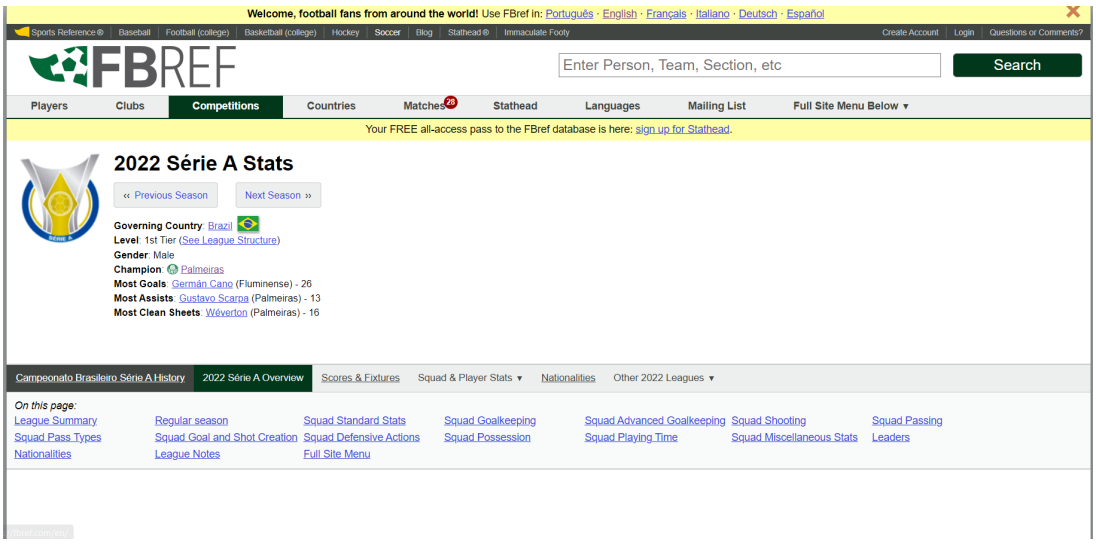
Pos	Equipe V·D·E	Pts	J	V	E	D	GP	GC	SG
1	Palmeiras **(C)**	**81**	38	23	12	3	66	27	+39
2	Internacional	**73**	38	20	13	5	58	31	+27
3	Fluminense	**70**	38	21	7	10	63	41	+22
4	Corinthians	**65**	38	18	11	9	44	36	+8
5	Flamengo	**62**	38	18	8	12	60	39	+21
6	Athletico Paranaense	**58**	38	16	10	12	48	48	0
7	Atlético Mineiro	**58**	38	15	13	10	45	37	+8
8	Fortaleza	**55**	38	15	10	13	46	39	+7
9	São Paulo	**54**	38	13	15	10	55	42	+13
10	América Mineiro	**53**	38	15	8	15	40	40	0
11	Botafogo	**53**	38	15	8	15	41	43	−2
12	Santos	**47**	38	12	11	15	44	41	+3
13	Goiás	**46**	38	11	13	14	40	53	−13
14	Red Bull Bragantino	**44**	38	11	11	16	49	59	−10
15	Coritiba	**42**	38	12	6	20	39	60	−21
16	Cuiabá	**41**	38	10	11	17	31	42	−11
17	Ceará	**37**	38	7	16	15	34	41	−7
18	Atlético Goianiense	**36**	38	8	12	18	39	57	−18
19	Avaí	**35**	38	9	8	21	34	60	−26
20	Juventude	**22**	38	3	13	22	29	69	−40

Source: Wikipedia, accessed on 07/10/2023

3.2 Data collection

The data analyzed was collected and organized in Excel spreadsheets, available at `FBref`[8], link to the page where the data was collected, as shown in Figure 3.

Figura 3: Home - Brazilian Championship 2022



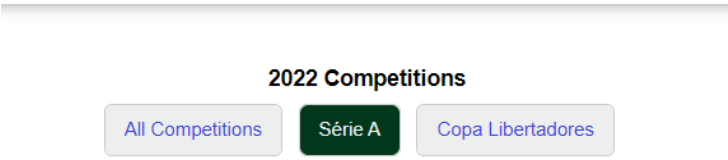
When you access this page, scroll down and you will find what is shown in Figure 4, make sure that the checkmark for the A series data is selected, as in Figure 5.

Figura 4: Club rankings

Rk	Squad	MP	W	D	L	GF	GA	GD	Pts	Pts/MP	xG	xGA	xGD	xGD/90	Attendance	Top Team Scorers	Goalkeeper	Notes
1	Palmeiras	38	23	12	3	66	27	+39	81	2.13	69.3	33.2	+36.1	+0.95	30,792	[illegible]	[illegible]	→ Copa Libertadores via league finish
2	Internacional	38	20	13	5	58	31	+27	73	1.92	55.8	45.1	+10.8	+0.28	19,889	Alexandre Alemão - 8	Daniel	→ Copa Sudamericana via league finish
3	Fluminense	38	21	7	10	63	41	+22	70	1.84	50.1	40.3	+9.7	+0.26	24,645	[illegible]	[illegible]	→ Copa Sudamericana via league finish
4	Corinthians	38	18	11	9	44	36	+8	65	1.71	40.9	41.0	-0.2	0.00	36,598	Rodrigo Garro - 10	Cássio	→ Copa Sudamericana via league finish
5	Flamengo	38	18	8	12	60	39	+21	62	1.63	65.4	35.8	+29.6	+0.78	45,630	Pedro, Gabriel Barbosa - 11	Santos	→ Copa Libertadores via league finish
6	Ath Paranaense	38	16	10	12	48	48	0	58	1.53	47.1	48.4	-1.3	-0.04	16,584	[illegible]	[illegible]	→ Copa Libertadores via league finish
7	Atlético Mineiro	38	15	13	10	45	37	+8	58	1.53	54.1	33.7	+20.4	+0.54	27,187	Hulk - 12	Everson	→ Copa Libertadores via league finish
8	Fortaleza	38	15	10	13	46	39	+7	55	1.45	44.5	42.9	+1.5	+0.04	27,488	Moises, Thiago Galhardo - 7	Fernando Miguel	→ Copa Libertadores via league finish
9	São Paulo	38	13	15	10	55	42	+13	54	1.42	49.7	44.4	+5.3	+0.14	27,157	Jonathan Calleri - 18	Jandrei	→ Copa Libertadores via league finish
10	América (MG)	38	15	8	15	40	40	0	53	1.39	45.6	49.7	-4.1	-0.11	2,876	Henrique Almeida - 6	Matheus Cavichioli	→ Copa Sudamericana via league finish
11	Botafogo (RJ)	38	15	8	15	41	43	-2	53	1.39	40.8	50.4	-9.6	-0.25	20,286	Erison - 7	Gatito Fernández	→ Copa Sudamericana via league finish
12	Santos	38	12	11	15	44	41	+3	47	1.24	46.2	48.0	-1.8	-0.05	9,575	Marcos Leonardo - 13	João Paulo	→ Copa Sudamericana via league finish
13	Goiás	38	11	13	14	40	53	-13	46	1.21	34.6	61.6	-27.0	-0.71	9,207	Pedro Raul - 19	Tadeu	
14	Bragantino	38	11	11	16	49	59	-10	44	1.16	52.7	49.5	+3.1	+0.08	4,219	Luan Cândido - 11	Cleiton Schwengber	
15	Coritiba	38	12	6	20	39	60	-21	42	1.11	42.3	53.6	-11.2	-0.30	15,495	Alef Manga - 9	Alex Muralha	
16	Cuiabá	38	10	11	17	31	42	-11	41	1.08	41.0	47.8	-6.8	-0.18	10,912	Deyverson - 6	Walter	
▼ 17	Ceará	38	7	16	15	34	41	-7	37	0.97	43.9	48.0	-4.0	-0.11	27,297	Stiven Mendoza - 10	João Ricardo Riedi	Relegated
▼ 18	Atl Goianiense	38	8	12	18	39	57	-18	36	0.95	43.1	51.2	-8.2	-0.21	5,911	Wellington Rato - 8	Ronaldo	Relegated
▼ 19	Avaí	38	9	8	21	34	60	-26	35	0.92	44.0	62.6	-18.6	-0.49	7,776	Guilherme Bissoli - 14	Vladimir	Relegated
▼ 20	Juventude	38	3	13	22	29	69	-40	22	0.58	33.4	57.1	-23.6	-0.62	3,200	Óscar Ruiz, Isidro Pitta - 5	César Reis	Relegated

[8] ⟨https://fbref.com/en/comps/24/2022/2022-Serie-A-Stats⟩, accessed on 17/09/2023

Figura 5: Correct data marking



3.2.1 Types of Variables

The spreadsheets collected consider different types of variables, some specific to goalkeepers, such as *Goalkeeping* and *Advanced Goalkeeping*, while the others are specific to line players: *Shooting*, with variables related to shooting; *Passing*, with variables related to passing; *Pass Types*, variables by type of pass; *Goal and Shot Creation*, a composition of variables measuring the creation of attacking moves; *Defensive Actions*, made up of defensive variables; *Possession*, organized by variables measuring the number of touches on the ball; *Playing Time*, made up of variables relating to playing time; and *Miscellaneous Stats*, with the other variables.

Each variable was coded with one or two letters followed by the number corresponding to its position in the database at the time of data collection: Gxx(*Goalkeeping* and *Advanced Goalkeeping*), Axx(*Shooting*), Pxx(*Passing*), TPxx(*Pass Types*), GSCxx(*Goal and Shot Creation*), Dxx(*Defensive Actions*), POxx(*Possession*), Txx(*Playing Time*) and Mxx(*Miscellaneous Stats*).

3.2.2 Data processing and division

To deal with the missing data, we chose to impute using the median. This choice was made considering that, for these specific variables, the data distribution was asymmetrical, making the median a more robust choice compared to the mean.

After organizing and standardizing the data, we proceeded with the general evaluation, considering the complete set of data, as well as the specific analyses for the first and last place finishers. The clubs were grouped into two groups: Group 1, containing the teams that qualified for the group stage of the Libertadores (Palmeiras, Internacional, Fluminense, Corinthians, Flamengo and Athetico Paranaense) and Group 2, the clubs in the last four places (Ceará, Atlético Goianiense, Avaí and Juventude). All the players who took part in the competition were considered.

- **Defenders**: General with 243 players, 72 in the G6 teams and 52 in the Z4 teams.
- **Midfielders**: Overall with 238 players, 61 in the G6 teams and 49 in the Z4 teams.
- **Forwards**: Overall with 221 players, 71 in the G6 teams and 46 in the Z4 teams.

The division by area of activity was due to the fact that, at the time of collection, the variable **"Pos"**, present in all the tables, classified players by sector and not position. Although the alternative of subdividing the data based on the most traditional positions was considered, the sources consulted did not show a consensus regarding the players' positions.

3.3 Defenders

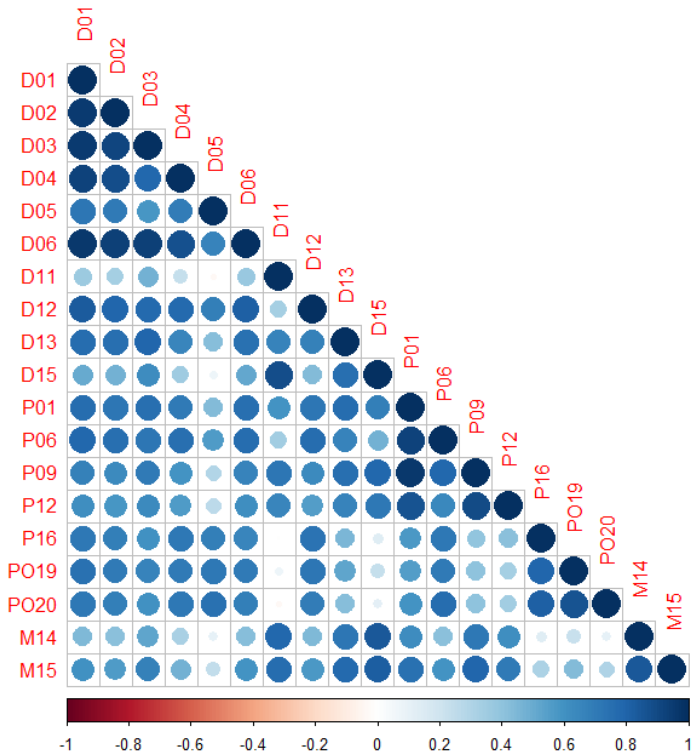
Made up of defenders and full-backs, the defensive sector plays an important role in the structure of a team. Defenders are responsible for protecting the penalty area and directly intercepting opposition attacks. A defender is expected to have good marking, anticipation of moves, a sharp aerial game and ball-winning skills. Full-backs, on the other hand, occupy a more external position and can contribute to both defense and attack. In addition to good marking, they are expected to be quick, have good passing and crossing skills, as well as being good going forward. To this end, a selection was made of the variables that would fit these requirements, as shown in Table 1.

Tabela 1: Description of defenders' variables - General

Name	Summary	Name	Summary
D01	Number of clean sheets	P01	Right passes
D02	Number of possessions won after a break	P06	Right short passes
D03	Splits in the defensive third from the field	P09	Right medium passes
D04	Splits in the central third from the field	P12	Right long passes
D05	Splits in the advanced third from the field	P16	Passes that generate a shot
D06	Individual duels won	P17	Passes in the final third
D11	Shot blocks	P19	Crosses into the penalty area
D12	Pass blocks	PO20	Domains in the attacking third
D13	Number of interceptions	M14	Aerial duels won
D15	Balls saved	M15	Aerial duels lost

Figure 6 shows the correlation matrix for all the players, in which you can see how each of the variables relates to the others. The color blue indicates a positive correlation, while shades of white indicate a non-existent relationship. D01, D02 and D03 appear to be strongly correlated. The variables for winning (M14) and losing (M15) aerial balls, as well as the passing variables, also appear to be strongly correlated. Another behavior that follows the same relationship as the others is between P19 and PO20, as well as between D11 and D15[9].

Figura 6: Correlation between variables - Defenders



3.3.1 Application of Principal Components

Principal Component Analysis was a useful tool for this study, especially for dealing with highly correlated variables. Multicollinearity occurs when there is a high correlation between variables, which can make it difficult to interpret the results. Choosing the Principal Component Analysis method allowed us to reduce the dimensionality of the data, thus creating new uncorrelated variables, which we call principal components. These components retain most of the original variation in the data, while being orthogonal to

[9]https://soccernovo.com/what-is-a-clearance-in-soccer/

each other, thus avoiding multicollinearity.

The eigenvalues measured the amount of variance retained in each component. It can be seen that the first eigenvalue is the largest, and the others gradually decrease. Eigenvalues greater than 1 indicate that the main component is responsible for more variability than would be accounted for by one of the original variables. As the data is standardized, a criterion was used as a cut-off point.

Tabela 2: Principal Component Analysis of defenders - General

Dimension	**Eigenvalue**	**% Cumulative Variance**
Dim.1	12.20557	64.23
Dim.2	3.450410	82.39
Dim.3	0.800312	86.61

Table 2 shows us the number of components, represented by "Dimension", the corresponding eigenvalues and the variance accumulated in each of them. The expected behavior is for the first component to retain most of the variability in the data. This behavior was observed in the table. The other components contribute gradually. As stated above, we are looking for components whose eigenvalue is above 1. In this case, the option of using two components, which together explain 82.61% of the variability present in the data, is appropriate.

3.3.2 Eigenvectors

The eigenvectors are associated with each of the components and represent the direction and magnitude of the changes in the original variables, where each eigenvector is a linear combination of these variables and describes the contribution of each one to that component. Each combination shows the contribution of the original variables.

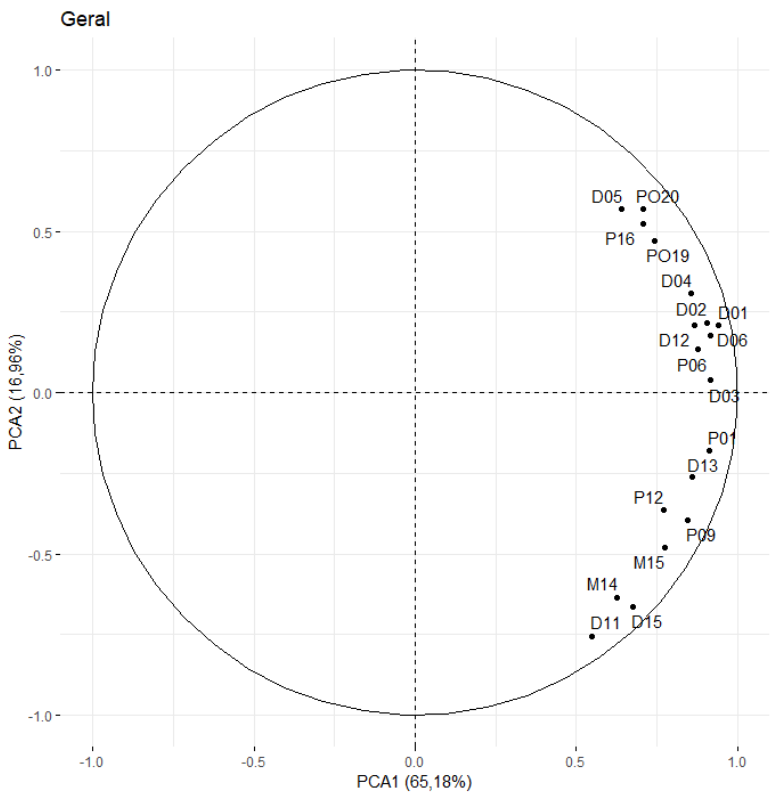
Tabela 3: Eigenvectors for CP1 and CP2 of the defenders

Variable	Average Performance (CP1))	Determining factors (CP2)
D01	0.2691	0.1122
D02	0.2590	0.1164
D03	0.2625	0.0206
D04	0.2449	0.1657
D05	0.1833	0.3057
D06	0.2620	0.0946
D11	0.1569	−0.4066
D12	0.2476	0.1127
D13	0.2455	−0.1401
D15	0.1936	−0.3573
P01	0.2605	−0.0974
P06	0.2509	0.0716
P09	0.2418	−0.2123
P12	0.2201	−0.1966
P16	0.2021	0.2810
PO19	0.2126	0.2527
PO20	0.2024	0.3064
M14	0.1791	−0.3415
M15	0.2211	−0.2580

In Table 3, now with the data reduced to two dimensions, we can see the eigenvectors associated with each variable, where each is represented in the two components. When considering a Cartesian plane, the direction of the variable must be interpreted from the sign of the eigenvector. If the eigenvector is positive in component 1 and component 2, the variable will be located in the 1st quadrant. Similarly, if both eigenvectors are negative, the variable will be in the 3rd quadrant. If the eigenvector is positive in component 1 and negative in component 2, the variable will be in the 4th quadrant; conversely, if it is negative in component 1 and positive in component 2, it will be in the 2nd quadrant. This interpretation shows the orientation of the variables in the first two principal components. It is important to note that variables with a negative sign indicate significant contrasts, highlighting the tactical and strategic priorities adopted in the current sporting scenario. This relationship can be seen in Figure 7.

Component 1 can be understood as a general average, representing common characteristics that all players share to a certain extent. On the other hand, Component 2, due to the contrasts observed in the coefficients, stands out for differentiating players in a more specific way. Variables such as D05, P16, PO19 and PO20 stand out in this sense, indicating that players who participated more in tackles in the central third, made passes that led to shots, crossed into the opponent's area and dominated in the attacking third stand out from the rest. This shows a greater importance for full-backs compared to defenders, with medium passes and some variables related to tackles appearing soon after.

Figura 7: Defenders' variables - General



3.3.3 Comparison between Groups

In order to see if there is a significant difference between the clubs in the top positions and the relegated clubs, we first analyzed both correlation matrices.

Figura 8: Defenders' correlations - G6

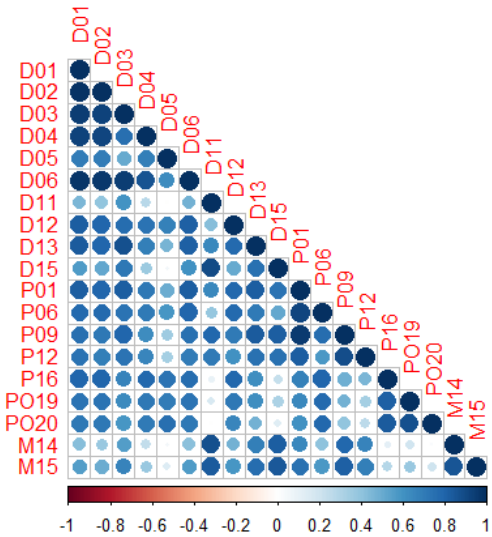
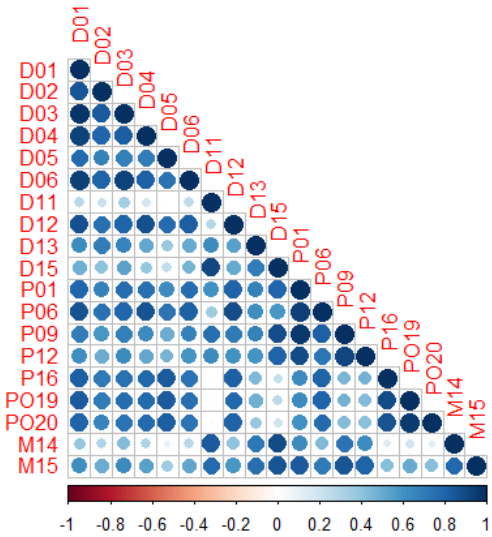


Figura 9: Defenders' correlations - Z4



What is immediately apparent from looking at Figures 8 and 9 is that the former has stronger correlations than the latter. For example, there is a strong correlation between the defensive variables. The passing variables also show the same behavior. This could also mean a difference in the importance of each variable in relation to the components.

Similarly to what was seen in Table 2, a Table4shows 2 Principal Components for both groups, capturing 84.84% and 84.72% of the variability in the data, respectively. This value is close to what was seen for the overall data.

Tabela 4: Principal Component Analysis of defenders - G6 and Z4

Dimension	Eigenvalue G6	% Variance G6	Eigenvalue Z4	% Variance Z4
Dim.1	12.66265	66.64	12.53262	65.96
Dim.2	3.457765	84.84	3.565422	84.72
Dim.3	0.724469	88.65	0.700576	88.41

The table shows 5 is similar to the first component of the general data. It is the second component of both that shows some differences between the groups. For Group 1, variables such as D11, D15 and those related to aerial duels are the ones that most indicate distinctive characteristics. This suggests that, in clubs qualifying for the Libertadores, players with higher scores in these variables stand out for shot blocking, saved balls and performance in aerial duels.

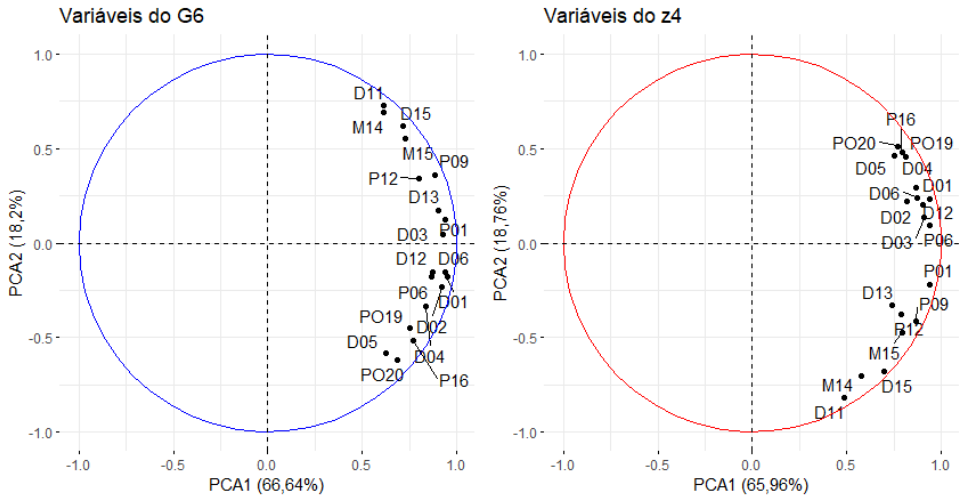
In contrast, Group 2 shows that variables such as P16, PO19, and PO20 are the most influential for Component 2. This indicates that, in relegated clubs, players with higher scores on these variables excel in tackles in the central third, passes that generate shots, crosses into the opponent's penalty area, and domination in the attacking third (which suggests that the active participation of the full-backs is more decisive than that of the defenders themselves).

Tabela 5: Eigenvectors related to G6 and Z4 - Defenders

Variable	Mean (G6)	Mean (Z4)	Determinants (G6)	Determinants (Z4)
D01	0.2669	0.2652	−0.0974	0.1246
D02	0.2592	0.2322	−0.1240	0.1173
D03	0.2612	0.2573	0.0250	0.0730
D04	0.2354	0.2443	−0.1790	0.1537
D05	0.1751	0.2129	−0.3130	0.2457
D06	0.2640	0.2470	−0.0827	0.1276
D11	0.1728	0.1378	0.3902	−0.4339
D12	0.2453	0.2551	−0.0812	0.1085
D13	0.2542	0.2089	0.0939	−0.1756
D15	0.2006	0.1972	0.3314	−0.3586
P01	0.2647	0.2647	0.0662	−0.1153
P06	0.2445	0.2656	−0.0972	0.0509
P09	0.2490	0.2441	0.1947	−0.2204
P12	0.2250	0.2227	0.1839	−0.2011
P16	0.2165	0.2237	−0.2790	0.2556
PO19	0.2117	0.2289	−0.2423	0.2425
PO20	0.1921	0.2185	−0.3333	0.2717
M14	0.1715	0.1626	0.3731	−0.3731
M15	0.2050	0.2248	0.2965	−0.2522

The relationship between the direction of the variables, explained in Table 5 can be better visualized in Figure 10.

Figura 10: Variáveis dos defensores - G6 e Z4



3.4 Midfielders

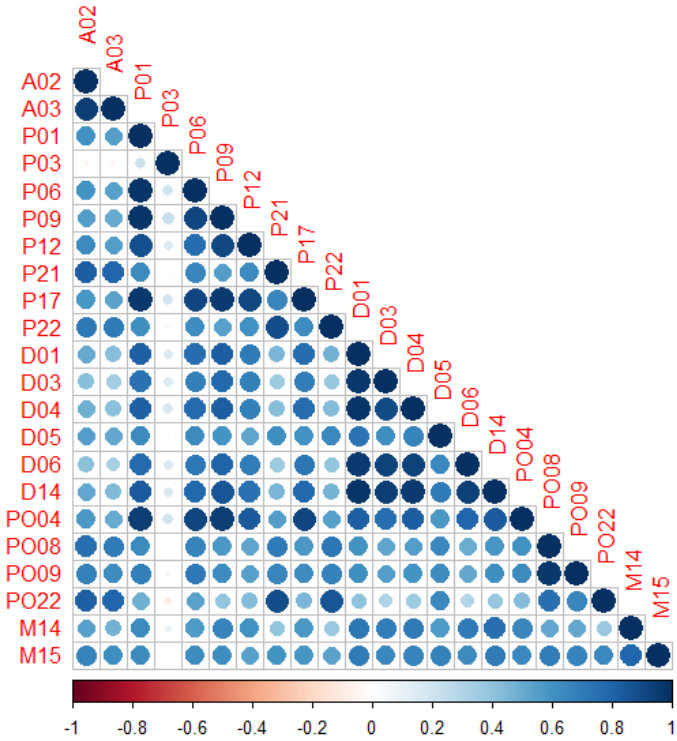
The characteristics of a good midfielder vary according to each team's formation. Midfielders can be divided into defensive midfielders and attacking midfielders. The main area of the pitch occupied by the former is in front of the defense. These players must be good at marking and recovering the ball, as well as being able to distribute play. The midfielders play an important role in protecting the defense and initiating play. Midfielders are the most creative part of the midfield. Their job is to distribute passes intelligently to leave their teammates in a position to finish off moves. They must also have the dribbling skills to overcome the opposition's marking. These players are also expected to have medium-distance finishing ability. Therefore, the variables that best fit these characteristics were chosen.

Tabela 6: Description of the half-campers' variables - General

Name	Summary	Name	Summary
A02	Finishing	D03 from the field	Splits in the defensive third
A03	Goal kicks	D04 of the field	Breakdowns in the central third
P01	Correct passes	D05	Individual duels won
P03	Passing success	D14	Number of interceptions
P06	Right short passes	D06	Individual duels won
P09	Right midfield passes	PO08	Dribble attempts
P12	Long passes right	PO09 successful	Dribbling attempts
P21	Passes that generate a shot including assists	P022	Passes that moved a ball towards goal
P17	Passes in the advanced third of the field	PO04	Touches in the central third
P22	Crosses + Complete passes into the penalty area	M14	Aerial duels won
D01	Number of tackles	M15	Aerial duels lost

Figure 11 shows the correlation between the variables mentioned above, and we can see that pass success does not show a positive correlation with any of the others. with any of the others. There is a stronger correlation between the finishing variables, Right passes with short and medium passes, as well as with long passes, the latter being a pair with a weaker correlation in relation to the others.

Figura 11: Correlation between variables - Half campers



3.4.1 Principal Component Analysis

In Table 7 one could choose to use 3 components, but the gain in variability from the second is only 5.5%, so to maintain the standard presented in the analysis for defenders we will use 2 components.

Tabela 7: Principal Component Analysis of campers - General

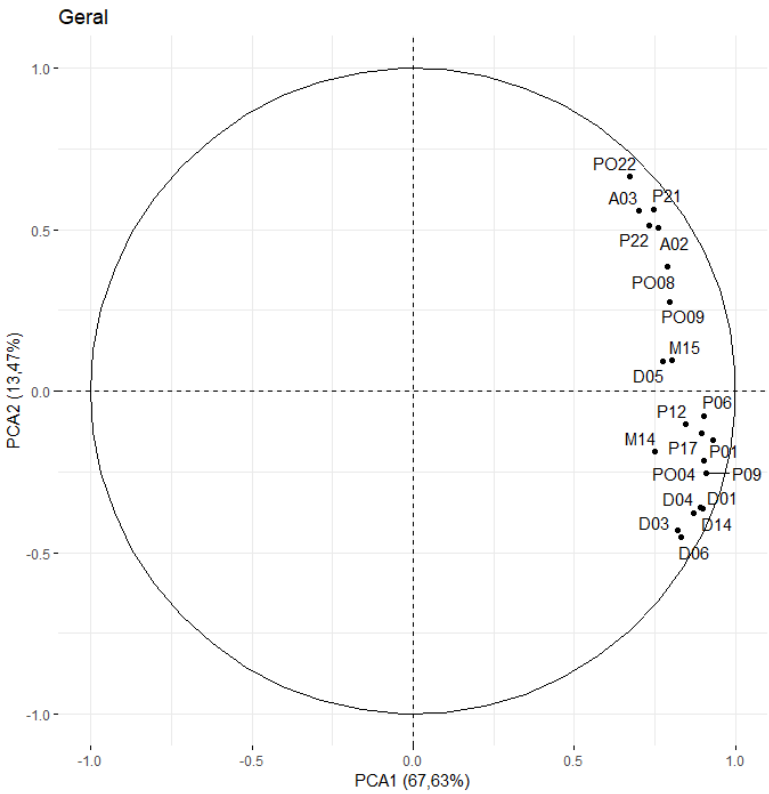
Dimension	Eigenvalue	% Cumulative Variance
Dim.1	14.20389	67.63
Dim.2	2.827185	81.10
Dim.3	1.155221	86.60
Dim.4	0.683667	89.85

As in Table 3, the first component shows a general average of the variables. Component 2, in turn, stands out for indicating the characteristics that most differentiate the players in the specific context of midfielders. Variables such as Shots, Shots on goal, Passes that generate a shot and Passes + Crosses into the penalty area, as well as Passes that move the ball towards the goal and Dribbling attempts show the quality of the midfielder as a determining factor.

Tabela 8: Eigenvectors for CP1 and CP2 of the midfielders

Variable	**Average Performance (CP1)**	**Determining factors (CP2)**
A02	0.2020	0.2999
A03	0.1852	0.3319
P01	0.2469	−0.0898
P06	0.2389	−0.0455
P09	0.2412	−0.1508
P12	0.2242	−0.0607
P21	0.1977	0.3352
P17	0.2369	−0.0769
P22	0.1939	0.3048
D01	0.2367	−0.2144
D03	0.2175	−0.2566
D04	0.2302	−0.2247
D05	0.2057	0.0547
D06	0.2202	−0.2694
D14	0.2386	−0.2170
PO04	0.2388	−0.1287
PO08	0.2095	0.2285
PO09	0.2106	0.1645
PO22	0.1778	0.3947
M14	0.1986	−0.1114
M15	0.2127	0.0569

Figura 12: Mid-campers' variables - General



3.4.2 Comparison between Groups

Continuing with the comparison between the groups, Figures 13 and 14 show the specific correlations in each set. The first figure shows similar patterns to those observed in the general context of all the clubs. However, the second figure highlights an extremely high correlation in the columns referring to Z4. A first assumption is that there was greater defensive participation by these players, and this may have contributed to this phenomenon, considering that most of the distinct correlations in the other figures involve defensive variables.

Figura 13: Midfielders' correlations - G6

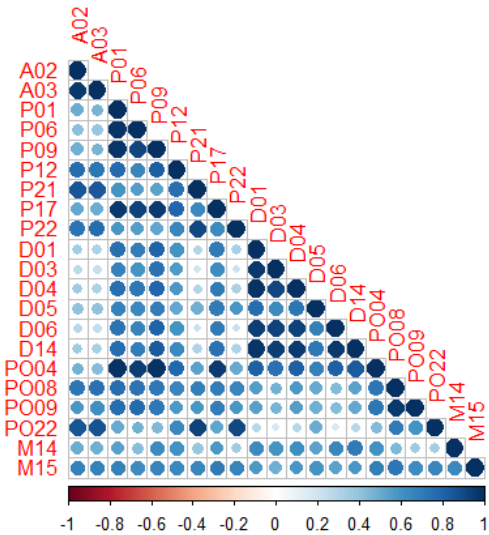


Figura 14: Midfielders' correlationss - Z4

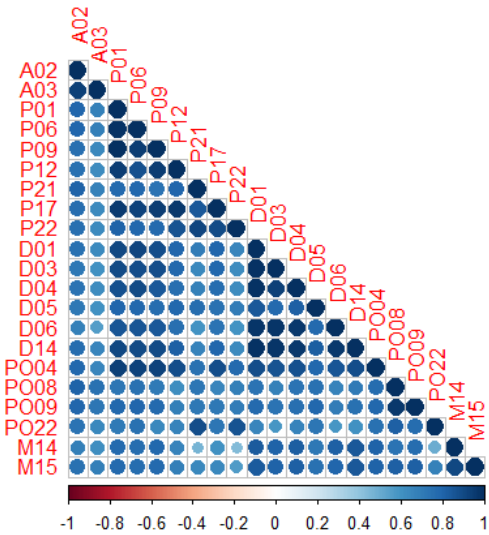


Table 9 shows a similar relationship when evaluating group 1 and the overall group. The first components retain 64.70% and 67.63%, respectively. Followed by a second component with more than 81% of variability. The components of group 2 show a different pattern. The first component explains more than 78% of the variability in the data. One factor that may have contributed to this was the high correlation seen in Figure 14.

Tabela 9: Principal Component Analysis of the midfielders - G6 e Z4

Dimension	Eingenvalue G6	% Variance G6	Eingenvalue Z4	% Variance Z4
Dim.1	13.58794	64.70	16.51925	78.66
Dim.2	3.546074	81.59	1.503625	85.82
Dim.3	1.196285	87.28	1.065126	90.89

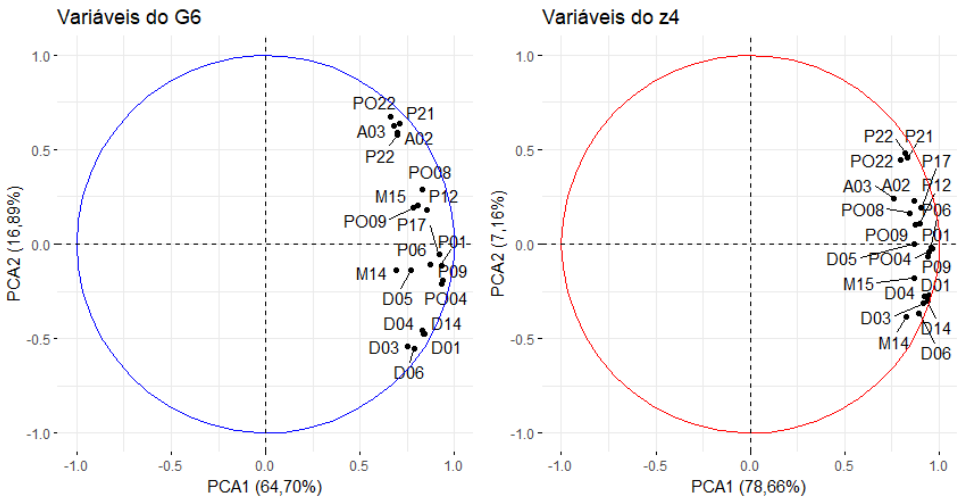
In Table 10, similarly to the previous tables, the first component shows the overall average of the variables. For component 2, the finishing variables have greater weight when we look at the first group, while the other variables have an equivalent contribution. The quality of the pass was also highlighted when we looked at variables P21, P22 and PO22. It is worth noting the difference in meaning of the variables P17 (Passes in the advanced third), D05 (Individual duels won) and M15 (Aerial duels lost).

Tabela 10: Eigenvectors related to G6 and Z4 - Meio campistas

Variable	Mean (G6)	Mean (Z4)	Determinants (G6)	Determinants (Z4)
A02	0.1883	0.2142	0.3146	0.1881
A03	0.1835	0.1872	0.3327	0.1974
P01	0.2518	0.2368	−0.0593	−0.0185
P06	0.2356	0.2357	−0.0560	−0.0134
P09	0.2535	0.2321	−0.1023	−0.0562
P12	0.2307	0.2208	0.0967	0.0904
P21	0.1924	0.2049	0.3400	0.3735
P17	0.2488	0.2226	−0.0294	0.1564
P22	0.1892	0.2013	0.3052	0.3946
D01	0.2266	0.2327	−0.2531	−0.2200
D03	0.2042	0.2254	−0.2879	−0.2570
D04	0.2248	0.2271	−0.2432	−0.2246
D05	0.2089	0.2141	−0.0746	0.0018
D06	0.2129	0.2200	−0.2950	−0.2987
D14	0.2289	0.2309	−0.2520	−0.2430
PO04	0.2532	0.2328	−0.1111	−0.0339
PO08	0.2246	0.2073	0.1539	0.1341
PO09	0.2126	0.2146	0.1006	0.0849
PO22	0.1798	0.1963	0.3591	0.3649
M14	0.1869	0.2033	−0.0729	−0.3161
M15	0.2186	0.2133	0.1083	−0.1466

This behavior becomes more evident when looking at Figure 15. It can be seen that there are no major variations in the direction of most of the variables. The variables mentioned above stand out: P17, M15 and D05, which show different directions in the graphs for component 2. It can be seen that P17 contributes more to group 2 in relation to group 1 and in general, M15 is more notable in group 1 than in group 2. And D05 is less important for group 1 compared to the others.

Figura 15: Midfielder variables - G6 e Z4



3.5 Forwards

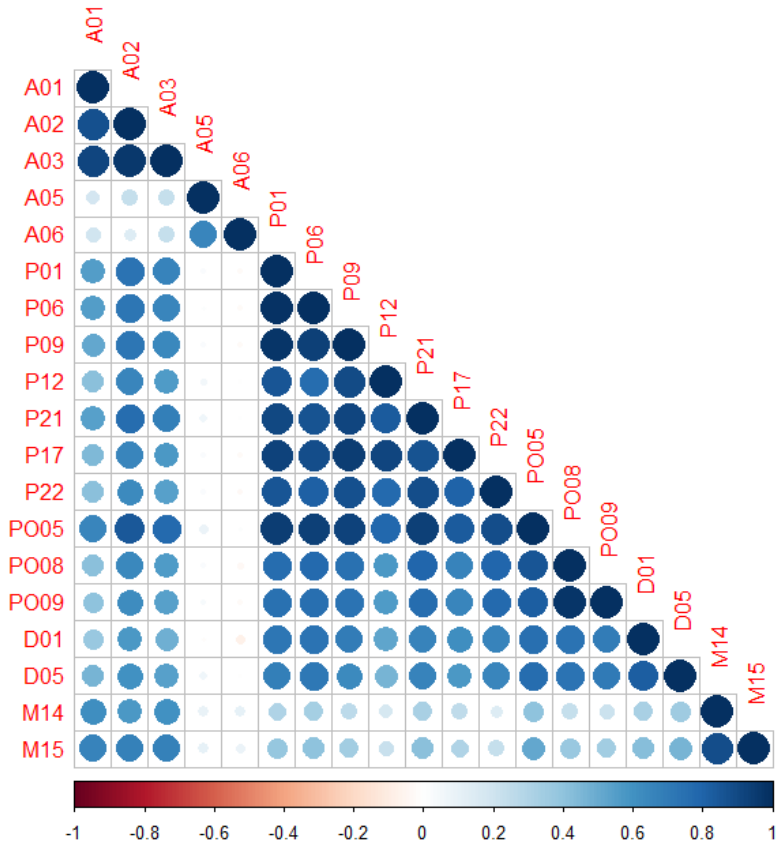
These are the players responsible for scoring goals. They play on the front line and their main function is to finish off moves. They can be divided into center forwards, who are the players who play more centrally, always looking to position themselves well to finish off moves, and winger, who play mainly on the sides of the pitch, are usually fast and skillful players, responsible for dribbling past opponents and creating goal-scoring situations.

Tabela 11: Description of fowards' variables - General

Name	Summary	Name	Summary
A01	Goals scored	P17	Passes in the advanced third field
A02	Finalizations	P22	Crosses + Complete passes into the penalty area
A03	Finishes on target	P005	Touches in the offensive third of the field offensive third of the field
A05	Average of shots per game	PO8	Dribbling attempts
A06	Average of shots right passes per game	PO09	Dribbling attempts successful
P01	Right passes	D01	Number of tackles
P06	Right short passes	D05	Individual duels won
P09	Medium passes correct	M14	Aerial duels won
P12	Long passes right	M15	Aerial duels lost
P21	Passes that lead to a shot		

Figure 16 shows that the averages of shots on goal and shots per game do not show even a moderate correlation with the other variables. This scenario can be considered an argument for removing these variables from the analysis. On the other hand, the variables related to attack seem to be more correlated with each other, although they show a moderate correlation with the others. In the context of passing, there is a high correlation between these variables and those of defense, contrasting with a lower correlation when compared to aerial exploitation.

Figura 16: Correlation between variables - Fowards



3.5.1 Principal Component Analysiss

Table 12 shows that the first 2 components explain more than 80% of the variability in the data, in line with what was seen for the previous groups. Therefore, the use of these 2 components was adopted.

Tabela 12: Principal Component Analysis of fowards - General

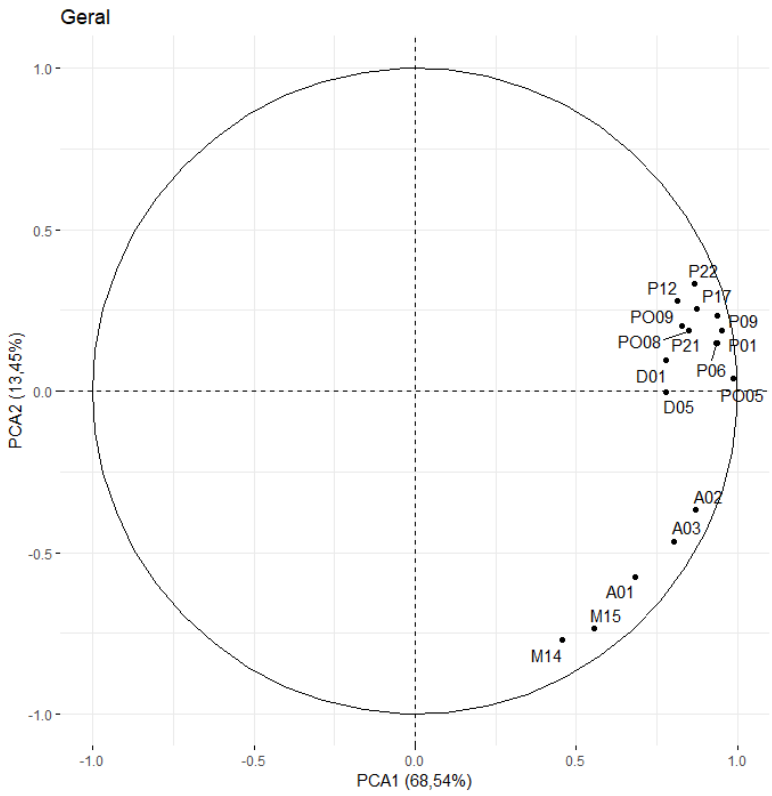
Dimension	Eigenvalue	% Variance
Dim.1	11.65245	68.54
Dim.2	2.287292	81.99
Dim.3	1.065049	88.26

Component 1, once again, indicates an average contribution from each variable. In component 2, the P22 variable stands out, initially indicating a greater participation by winger players in crosses and complete passes into the penalty area, as the A01, A02 and A03 variables indicate a contrast in relation to the aforementioned variable. Other passing variables also made a significant contribution, such as P17 and P09. Variables showing performance in aerial duels show us that these elements have negative values, indicating an inverse association.

Tabela 13: Autovetores para CP1 e CP2 dos atacantes

Variable	**Average Performance (CP1)**	**Determining factors (CP2)**
A01	0.1994	−0.3808
A02	0.2544	−0.2425
A03	0.2346	−0.3075
P01	0.2783	0.1238
P06	0.2741	0.0978
P09	0.2739	0.1545
P12	0.2377	0.1839
P21	0.2737	0.0975
P17	0.2559	0.1677
P22	0.2539	0.2190
PO05	0.2889	0.0257
PO08	0.2484	0.1240
PO09	0.2420	0.1323
D01	0.2277	0.0621
D05	0.2279	−0.0022
M14	0.1330	−0.5101
M15	0.1623	−0.4856

Figura 17: Forward variables - General



3.5.2 Comparison between groups

What can be seen in Figures 18 and 19 is a high level of interaction between the variables of performance in finishing, with the highest correlation between goals and finishing in the first group standing out. There is also a strong relationship between attacking participation and passing in general, showing that a good striker must also construct moves. On the other hand, only aerial performance doesn't seem to be highly correlated with the others.

Figura 18: Forward correlations - G6

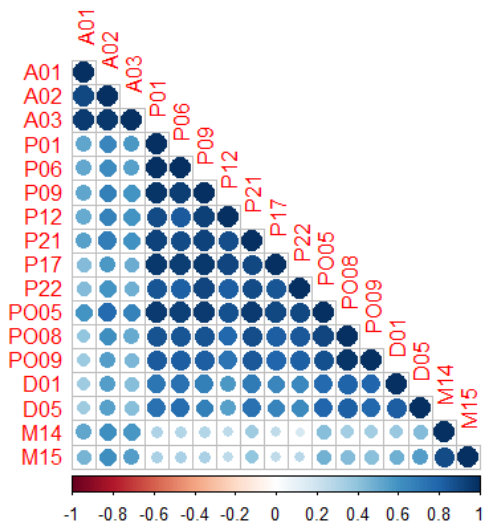


Figura 19: Forward correlations - Z4

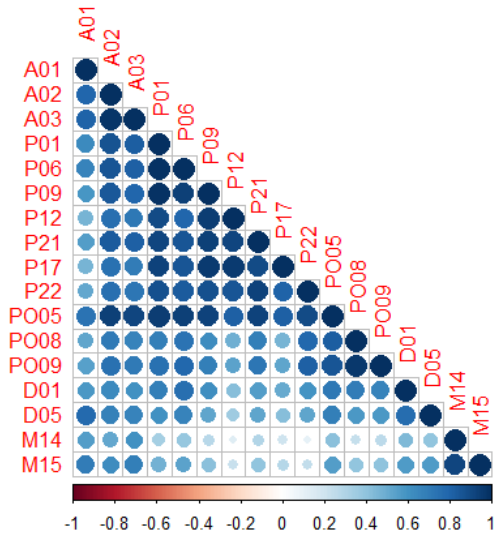


Table 14 indicates a similar behavior to that considered for all players. The first two components explain more than 80% of the variability in the data, as also seen in Table 12.

Tabela 14: Principal Component Analysis of the fowards - G6 e Z4

Dimension	Eigenvalue G6	% Variance G6	Eigenvalue Z4	% Variance Z4
Dim.1	11.89517	69.97	11.93968	70.23
Dim.2	2.283106	83.40	2.177252	83.04
Dim.3	1.268185	90.86	1.141576	89.75

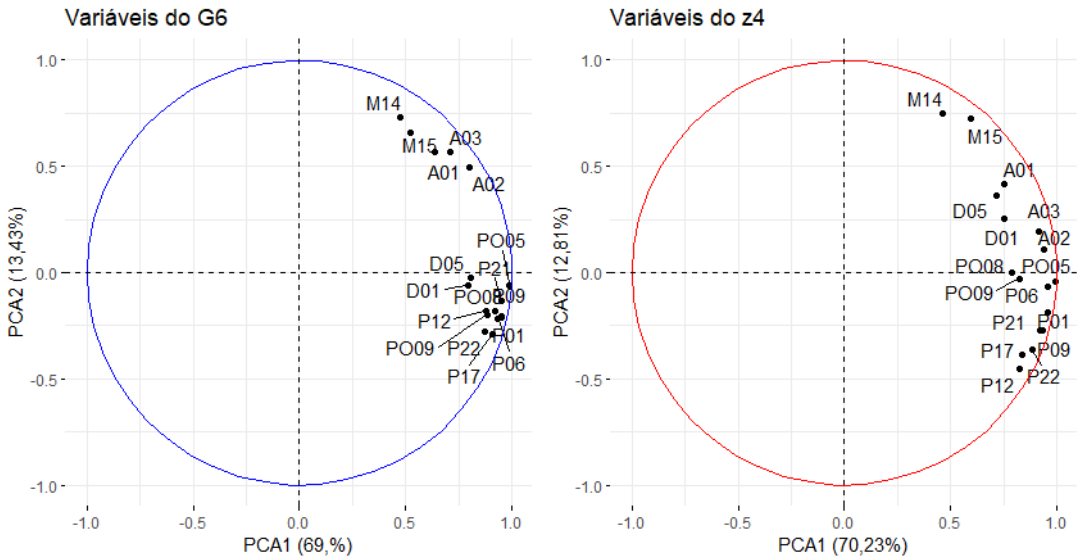
Table 15 shows intriguing patterns in component 2: the players in Group 6 show a notable superiority in offensive aspects, particularly in the variables A01, A02 and A03, when compared to the players in Group Z4. Conversely, the players in Group Z4 display a more prominent defensive participation compared to the first group, suggesting a significant contribution in the defensive moments of matches. Both groups show substantial involvement in aerial performance, indicating a notable presence in this aspect of the game.

Tabela 15: Autovectors related to G6 and Z4 - Fowards

Variable	Mean (G6)	Mean (Z4)	Determinants (G6)	Determinants (Z4)
A01	0.1854	0.2185	0.3773	0.2830
A02	0.2325	0.2719	0.3296	0.0750
A03	0.2064	0.2654	0.3753	0.1324
P01	0.2761	0.2780	−0.1407	−0.1273
P06	0.2709	0.2765	−0.1429	−0.0459
P09	0.2759	0.2698	−0.1371	−0.1861
P12	0.2555	0.2384	−0.1185	−0.3076
P21	0.2754	0.2664	−0.0891	−0.1827
P17	0.2633	0.2424	−0.1921	−0.2600
P22	0.2534	0.2555	−0.1838	−0.2450
PO05	0.2871	0.2869	−0.0385	−0.0273
PO08	0.2665	0.2283	−0.1202	−0.0002
PO09	0.2563	0.2381	−0.1306	−0.0184
D01	0.2302	0.2183	−0.0388	0.1718
D05	0.2335	0.2075	−0.0156	0.2459
M14	0.1384	0.1334	0.4822	0.5067
M15	0.1510	0.1728	0.4351	0.4903

The relationship between the attacking and defending variables is best expressed in Figure 20.

Figura 20: Foward's variables - G6 e Z4



3.6 Creating scores

The introduction of Principal Component Analysis (PCA) has emerged as a valuable approach to investigating the underlying structure of the data. The scores resulting from this analysis provide a representative synthesis of the relationships between orthogonal variables, enabling a simplified interpretation of the performance of groups in different positions. In this section, I will highlight the top 10 scores for each component, to improve the focus of the study.

The scores play a fundamental role after the reduction from P to k dimensions. In this context, the k principal components emerge as new individuals, outlining the basis for subsequent analyses. The use of these component scores permeates the entire analysis, providing a deeper understanding of the structure of the data and facilitating the interpretation of significant relationships between variables, thus refining the perception of the groups' performance.

3.6.1 Defenders

Table 5 displayed the eigenvectors associated with each new variable, which will play a crucial role in calculating the expression corresponding to each component. These eigenvectors, derived during the analysis process, provided the directions and magnitudes that guide the contribution of each variable to the formation of the principal components. We therefore have:

$$CP1_{G6} = 0.2669X_{D01} + 0.2592X_{D02} + \ldots + 0.2050X_{M15}$$

$$CP1_{Z4} = 0.2652X_{D01} + 0.2322X_{D02} + \ldots + 0.2248X_{M15}$$

$$CP2_{G6} = -0.0974X_{D01} - 0.1240X_{D02} + \ldots + 0.2965X_{M15}$$

$$CP2_{Z4} = 0.1246X_{D01} + 0.1173X_{D02} + \ldots - 0.2522X_{M15}$$

Tabela 16: TOP 10 Best Defenders (CP1) - G6

Name	**Team**	**Score**
Marcos Rocha	Palmeiras	8.8736
Renê	Internacional	8.6451
Ayrton Lucas	Flamengo	6.7844
Murilo	Palmeiras	5.9682
Joaquín Piquerez	Palmeiras	5.9470
Gustavo Gómez	Palmeiras	5.5760
Matheuzinho	Flamengo	5.4627
Fabricio Bustos	Internacional	5.3991
Samuel Xavier	Fluminense	5.2889
Pedro Henrique	Athlético Paranaense	4.8267

Tabela 17: TOP 10 Best Defenders (CP1) - Z4

Name	Team	Score
Kevin Avaí	Avaí	9.5042
Bruno Cortez	Avaí	7.0032
Nino Paraíba	Ceará	6.9274
Rodrigo Alves Soares	Juventude	6.8536
Jefferson	Atlético Goianiense	6.2297
Luis Eduardo	Atlético Goianiense	4.4307
Paulo Henrique	Juventude	3.8991
Luiz Otávio	Ceará	3.7370
Wanderson	Atlético Goianiense	3.0199
Rafael Forster	Juventude	2.6746

Tabela 18: TOP 10 Best Defenders (CP2) - G6

Name	Team	Score
Gustavo Gómez	Palmeiras	5.2002
Vitão	Internacional	4.9892
Murilo	Palmeiras	4.1358
Pedro Henrique	Athlético Paranaense	3.6069
Gil	Corinthians	2.9372
Manoel	Fluminense	2.8321
Pablo	Flamengo	2.7796
Nino	Fluminense	2.6996
Gabriel Mercado	Internacional	2.5342
Rodrigo Moledo	Internacional	2.0810

Tabela 19: TOP 10 Best Defenders (CP2) - Z4

Name	Team	Score
Paulo Henrique	Juventude	3.7366
Nino Paraíba	Ceará	2.9969
Natanael	Avaí	2.4374
Bruno Cortez	Avaí	2.3742
Arthur Henrique	Atlético Goianiense	2.3547
Kevin	Avaí	1.8630
Luis Eduardo	Atlético Goianiense	1.8135
Hayner	Atlético Goianiense	1.4352
Rodrigo Alves Soares	Juventude	1.4348
Renato	Avaí	1.2976

3.6.2 Midfielders

Table 10 shows the following components.

$$CP1_{G6} = 0.1883X_{A02} + 0.1835X_{A03} + \ldots + 0.2186X_{M15}$$

$$CP1_{Z4} = 0.2142X_{A02} + 0.1872X_{A03} + \ldots + 0.2133X_{M15}$$

$$CP2_{G6} = 0.3146X_{A02} + 0.3327X_{A03} + \ldots + 0.1083X_{M15}$$

$$CP2_{Z4} = 0.1881X_{A02} + 0.1974X_{A03} + \ldots - 0.1466X_{M15}$$

Tabela 20: TOP 10 Best Midfielders (CP1) - G6

Name	Team	Score
André	Fluminense	8.7023
Carlos de Pena	Internacional	7.0393
Gustavo Scarpa	Palmeiras	6.7231
Ganso	Fluminense	6.5023
Danilo	Palmeiras	6.3476
Zé Rafael	Palmeiras	6.2297
Gabriel	Internacional	5.6923
Du Queiroz	Corinthians	5.0596
João Gomes	Flamengo	5.0558
Everton Ribeiro	Flamengo	5.0373

Tabela 21: TOP 10 Best Midfielders (CP1) - Z4

Name	Team	Score
Raniele	Avaí	9.3338
Marlon Freitas	Atlético Goianiense	8.6737
Bruno Silva	Avaí	8.2286
Richard	Ceará	8.0136
Jadson	Juventude	7.7132
Chico	Juventude	7.0270
Gabriel Baralhas	Atlético Goianiense	6.5382
Jorginho	Atlético Goianiense	6.2022
Eduardo de Biasi	Avaí	5.1835
Richardson	Ceará	4.9426

Tabela 22: TOP 10 Best Midfielders (CP2) - G6

Name	Team	Score
Gustavo Scarpa	Palmeiras	8.4592
David Terans	Atlético Paranaense	3.9961
Alan Patrick	Internacional	3.3883
Giorgian De Arrascaeta	Flamengo	3.2656
Everton Ribeiro	Flamengo	1.9743
Raphael Veiga	Palmeiras	1.6362
Vitor Bueno	Atlético Paranaense	1.3763
Nathan	Fluminense	1.3017
Taison	Internacional	1.1268
Eduard Atuesta	Palmeiras	1.0562

Tabela 23: TOP 10 Best Midfielders (CP2) - Z4

Name	Team	Score
Jorginho	Atlético Goianiense	3.3626
Marlon Freitas	Atlético Goianiense	2.7215
Chico	Juventude	2.0471
Felipe Pires	Juventude	1.5735
Yuri Oliveira Lima	Juventude	1.4864
Kelvin	Atlético Goianiense	1.4594
Eduardo de Biasi	Avaí	1.4288
Jadson	Juventude	1.2526
Bruno Nazário	Juventude	1.1635
Jean Pyerre	Avaí	1.0469

3.6.3 Fowards

Finally, Table 15 shows the following components.

$$CP1_{G6} = 0.1854X_{A01} + 0.2325X_{A02} + \ldots + 0.1510X_{M15}$$

$$CP1_{Z4} = 0.2185X_{A01} + 0.2719X_{A02} + \ldots + 0.1728X_{M15}$$

$$CP2_{G6} = 0.3773X_{A01} + 0.3296X_{A02} + \ldots + 0.4351X_{M15}$$

$$CP2_{Z4} = 0.2830X_{A01} + 0.0750X_{A02} + \ldots + 0.4903X_{M15}$$

Tabela 24: TOP 10 Best fowards (CP1) - G6

Name	Team	Score
Jhon Arias	Fluminense	13.6314
Dudu	Palmeiras	10.1673
Roger Guedes	Corinthians	7.4178
Germán Cano	Fluminense	4.9431
Gabriel Barbosa	Flamengo	4.8527
Alexandre Alemão	Internacional	4.7802
Tomás Cuello	Athletico Paranaense	4.6934
Matheus Martins	Fluminense	4.6211
Everton Soares	Flamengo	4.4570
Adson	Corinthians	3.7517

Tabela 25: TOP 10 Best fowards (CP1) - Z4

Name	Team	Score
Wellington Rato	Atlético Goianiense	9.4648
Vinícius	Ceará	8.4954
Stiven Mendoza	Ceará	6.8942
William Pottker	Avaí	5.5329
Lima	Ceará	5.5087
Diego Churín	Atlético Goianiense	4.7605
Guilherme Bissoli	Avaí	4.5207
Shaylon	Atlético Goianiense	3.4554
Airton	Atlético Goianiense	3.2583
Isidro Pitta	Juventude	3.2103

Tabela 26: TOP 10 Best fowards (CP2) - G6

Name	Team	Score
Germán Cano	Fluminense	5.7478
Rony	Palmeiras	4.4828
Alexandre Alemão	Internacional	3.4286
Pablo	Athlético Paranaense	3.2534
Yuri Alberto	Corinthians	3.2015
Pedro	Flamengo	2.8252
Vitor Roque	Athlético Paranaense	2.6024
Rômulo	Athlético Paranaense	1.4776
Rafael Navarro	Palmeiras	1.4507
Pedro Henrique	Internacional	1.2449

Tabela 27: TOP 10 Best fowards (CP2) - Z4

Name	Team	Score
Diego Churín	Atlético Goianiense	5.1004
Guilherme Bissoli	Avaí	3.6243
Cléber	Ceará	2.6824
Isidro Pitta	Juventude	2.0475
Vitor Gabriel	Juventude	1.8893
Ricardo Bueno	Juventude	1.3170
Zé Roberto	Ceará	0.9361
Stiven Mendoza	Ceará	0.8834
Matheus Peixoto	Ceará	0.7293
Airton	Atlético Goianiense	0.5649

4 Conclusions

The analysis applied to data from the 2022 Brazilian Championship was used to investigate the underlying structure in player performance. The data was segmented into three distinct groups, subsequently subdivided according to classification. The choice of 2 components proved effective in evaluating the eigenvalues and quantifying the variability retained by them. In addition, the generation of scores provided a clearer view of the real performance of each of the top 10 players in each component.

The analysis of the defenders provided valuable insights into the relationship between the attacking presence variables, suggesting a more active role for the full-backs. This finding highlights the importance of a good full-back not only in defense, but also in the ability to distribute and construct plays, highlighting the versatility required of these players. Furthermore, when examining Component 2, it was observed that the G6 players showed an even better defensive performance than the Z4 players. This difference suggests that the defenders of the clubs that qualified directly for the Libertadores played a more prominent role than the second group. This finding reinforces the crucial importance of a solid defense not only to protect the goal, but also to build the foundations of a team's success, indicating that a good defense is not only limited to preventing goals, but also plays a key role in building the path to victory.

The midfielders revealed some discrepancies when we compared the directions of some variables in the two groups, especially in the context of the second component. However, a common point identified in both groups was the emphasis on the quality of passing and the value placed on finishing ability. This observation reflects the fundamental idea that a quality midfielder should primarily focus on creating moves, while, whenever possible, demonstrating effective finishing skills

As for the strikers, we observed some disparities, especially in component 2, which highlighted superior attacking quality on the part of Group 1 players. This suggests that these strikers were more efficient in this particular aspect. On the other hand, the players in Group 2, in component 2, showed a more prominent participation in defensive situations. This observation suggests, for example, a qualitative superiority of the G6 strikers compared to the second group.

Finally, by using the scores, you can see how the players scored in each component, thus simplifying the assessment of their overall performance. This approach provides a clearer and more objective view of the players' individual contributions to the different

aspects of the game.

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